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Optimizing Intent Matching In Contact Center

Abstract

Respective confidence scores for a query with respect to respective intents are determined. An indication associated with the query is provided to a client device for review based on a determination that the query does not match any of the respective intents based on the respective confidence scores. Data representing an intent of the respective intents that matches the query is received from the client device. An intent matching engine is trained to associate the query with the intent based on the received data.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation of U.S. application Ser. No. 18/191,271, filed Mar. 28, 2023, which is a continuation of International Application Number PCT/CN2023/073011, filed Jan. 18, 2023, the entire disclosures of which are hereby incorporated by reference.

FIELD

[0002] This disclosure relates to training an intent matching engine of a contact center which may be implemented over a telephone or computer network.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] This disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to-scale. On the contrary, the dimensions of the various features are arbitrarily expanded or reduced for clarity.

[0004] FIG. 1 is a block diagram of an example of an electronic computing and communications system.

[0005] FIG. 2 is a block diagram of an example internal configuration of a computing device of an electronic computing and communications system.

[0006] FIG. 3 is a block diagram of an example of a software platform implemented by an electronic computing and communications system.

[0007] FIG. 4 is a block diagram of an example of a contact center system.

[0008] FIG. 5 is a block diagram of an example of a system for training an intent matching engine.

[0009] FIG. 6 is a block diagram of an example of stored data associated with a query.

[0010] FIG. 7 is a number line of values of confidence scores.

[0011] FIG. 8 is an example graphical user interface (GUI) for matching intents to queries at a client device.

[0012] FIG. 9 is a flowchart of an example of a technique for training an intent matching engine of a contact center.

DETAILED DESCRIPTION

[0013] The use of contact centers by or for service providers is becoming increasingly common to address customer support requests over various modalities, including telephony, video, text messaging, chat, and social media. In one example, a contact center may be implemented by an operator of a software platform, such as a unified communications as a service (UCaaS) platform, for a customer of the operator. Users of the customer may engage with the contact center to address support requests over one or more communication modalities enabled for use with the contact center by the software platform. In another example, the operator of such a software platform may implement a contact center to address customer support requests related to the software platform itself.

[0014] A contact center may use a server to process a contact center engagement from a user device. In some implementations, upon a connection of the user device to the server, the server prompts a user of the user device to provide (e.g., by typing or speaking) a query indicating the reason for initiating the contact center engagement. The query is processed by an intent matching engine at the server, which leverages artificial intelligence techniques (e.g., a deep convolutional neural network) to match the query to an intent. The intent is a member of a predefined set of intents representing what the user wishes to accomplish. The intent may be associated with at least

one of a workflow, a set of agents, or an automatic response. For example, a contact center of a bank may have an intent “close account,” with all queries that lead to the request to close an account being associated with a common workflow (e.g., prompting the user to verify their identity and closing the account upon completing the verification), a set of agent responsible for closing accounts, or an automatic response (e.g., including a link to a webpage explaining how to close an account).

[0015] To train the intent matching engine, a supervised learning technique with training data including manually generated (e.g., by an administrator of a business using the contact center) queries and associated intents may be used. Training the intent matching engine with queries and intents similar to queries and intents that occur in the real world may be useful because this training data most resembles the real world inference data on which the intent matching engine is to be used, and reflects the latest changes in queries submitted by users (e.g., due to user circumstances or needs changing over time). However, it may be costly in person-hours to generate the training data, as it might take many person-hours to generate a large number of queries and to specify the intents that should be associated with the generated queries. More efficient approaches for training the intent matching engine may be desirable.

[0016] Implementations of this disclosure address problems such as these by generating a training data set for the intent matching engine based on a manual review of a subset of the queries processed by the intent matching engine, and automatically selecting the subset of the queries for the manual review. A contact center server accesses queries provided by user devices. The contact center server determines a confidence score for one or more of the accessed queries using an intent matching engine. The confidence score represents a likelihood that a query matches to a given intent. For example, the query, “I wish to close my credit card account,” might match to the intent “close account” with a confidence score of 0.96, indicating that the server is highly confident that “close account” is the user's intent. If the confidence score for matching a given query to a given intent exceeds a threshold, the intent matching engine matches the given query to the given intent.

[0017] The contact center server generates a set of queries that are not matched to any intent using the above technique. From this set of queries, the contact center server selects a subset for use in further training (e.g., via online learning) the intent matching engine. The subset may be selected based on the confidence scores or the timestamps of the queries in the set. The confidence score may be useful to identify queries that are “almost” matched to intents (e.g., having a confidence score of 0.76, where the threshold for matching to the intent is 0.8). It may be beneficial to have the intent matching engine trained on these queries that were “almost” matched to the intent because, if the intent matching engine leverages an artificial neural network, only minor modifications of some of the weights applied in the neural network may be needed to increase the confidence score above the threshold. The timestamp may be useful because the structure of queries (and intents associated with the queries) may change over time in response to new product launches or changes in customer needs. For example, if a company that uses a contact center launches a new product line, customers of the company may access the contact center to purchase, ask questions about, or troubleshoot products associated with the new product line. The timestamp may be used to track queries and intents and changes therein over time. The contact center server transmits the subset of queries to one or more administrator devices. In response, users of the administrator devices manually match queries in the subset to intents to generate training data. The training data is received at the contact center server and used to further train the intent matching engine, for example, using online learning techniques.

[0018] As used herein, the term “engine” may include, among other things, a component of a physical computer or virtual machine that performs certain functions. The engine may be implemented using software that is stored in a memory and executed by processing circuitry. Alternatively, the engine may be hard-wired into the processing circuitry and implemented using hardware. In some cases, the engine may include both software and hardware components. An

engine may include one or more sub-engines, each of which performs part of the functionality of the engine.

[0019] To describe some implementations in greater detail, reference is first made to examples of hardware and software structures used to implement training an intent matching engine of a contact center. FIG. 1 is a block diagram of an example of an electronic computing and communications system **100**, which can be or include a distributed computing system (e.g., a client-server computing system), a cloud computing system, a clustered computing system, or the like.

[0020] The system **100** includes one or more customers, such as customers **102A** through **102B**, which may each be a public entity, private entity, or another corporate entity or individual that purchases or otherwise uses software services, such as of a UCaaS platform provider. Each customer can include one or more clients. For example, as shown and without limitation, the customer **102A** can include clients **104A** through **104B**, and the customer **102B** can include clients **104C** through **104D**. A customer can include a customer network or domain. For example, and without limitation, the clients **104A** through **104B** can be associated or communicate with a customer network or domain for the customer **102A** and the clients **104C** through **104D** can be associated or communicate with a customer network or domain for the customer **102B**.

[0021] A client, such as one of the clients **104A** through **104D**, may be or otherwise refer to one or both of a client device or a client application. Where a client is or refers to a client device, the client can comprise a computing system, which can include one or more computing devices, such as a mobile phone, a tablet computer, a laptop computer, a notebook computer, a desktop computer, or another suitable computing device or combination of computing devices. Where a client instead is or refers to a client application, the client can be an instance of software running on a customer device (e.g., a client device or another device). In some implementations, a client can be implemented as a single physical unit or as a combination of physical units. In some implementations, a single physical unit can include multiple clients.

[0022] The system **100** can include a number of customers and/or clients or can have a configuration of customers or clients different from that generally illustrated in FIG. 1. For example, and without limitation, the system **100** can include hundreds or thousands of customers, and at least some of the customers can include or be associated with a number of clients.

[0023] The system **100** includes a datacenter **106**, which may include one or more servers. The datacenter **106** can represent a geographic location, which can include a facility, where the one or more servers are located. The system **100** can include a number of datacenters and servers or can include a configuration of datacenters and servers different from that generally illustrated in FIG. 1. For example, and without limitation, the system **100** can include tens of datacenters, and at least some of the datacenters can include hundreds or another suitable number of servers. In some implementations, the datacenter **106** can be associated or communicate with one or more datacenter networks or domains, which can include domains other than the customer domains for the customers **102A** through **102B**.

[0024] The datacenter **106** includes servers used for implementing software services of a UCaaS platform. The datacenter **106** as generally illustrated includes an application server **108**, a database server **110**, and a telephony server **112**. The servers **108** through **112** can each be a computing system, which can include one or more computing devices, such as a desktop computer, a server computer, or another computer capable of operating as a server, or a combination thereof. A suitable number of each of the servers **108** through **112** can be implemented at the datacenter **106**. The UCaaS platform uses a multi-tenant architecture in which installations or instantiations of the servers **108** through **112** is shared amongst the customers **102A** through **102B**.

[0025] In some implementations, one or more of the servers **108** through **112** can be a non-hardware server implemented on a physical device, such as a hardware server. In some implementations, a combination of two or more of the application server **108**, the database server **110**, and the telephony server **112** can be implemented as a single hardware server or as a single

non-hardware server implemented on a single hardware server. In some implementations, the datacenter **106** can include servers other than or in addition to the servers **108** through **112**, for example, a media server, a proxy server, or a web server.

[0026] The application server **108** runs web-based software services deliverable to a client, such as one of the clients **104A** through **104D**. As described above, the software services may be of a UCaaS platform. For example, the application server **108** can implement all or a portion of a UCaaS platform, including conferencing software, messaging software, and/or other intra-party or inter-party communications software. The application server **108** may, for example, be or include a unitary Java Virtual Machine (JVM).

[0027] In some implementations, the application server **108** can include an application node, which can be a process executed on the application server **108**. For example, and without limitation, the application node can be executed in order to deliver software services to a client, such as one of the clients **104A** through **104D**, as part of a software application. The application node can be implemented using processing threads, virtual machine instantiations, or other computing features of the application server **108**. In some such implementations, the application server **108** can include a suitable number of application nodes, depending upon a system load or other characteristics associated with the application server **108**. For example, and without limitation, the application server **108** can include two or more nodes forming a node cluster. In some such implementations, the application nodes implemented on a single application server **108** can run on different hardware servers.

[0028] The database server **110** stores, manages, or otherwise provides data for delivering software services of the application server **108** to a client, such as one of the clients **104A** through **104D**. In particular, the database server **110** may implement one or more databases, tables, or other information sources suitable for use with a software application implemented using the application server **108**. The database server **110** may include a data storage unit accessible by software executed on the application server **108**. A database implemented by the database server **110** may be a relational database management system (RDBMS), an object database, an XML database, a configuration management database (CMDB), a management information base (MIB), one or more flat files, other suitable non-transient storage mechanisms, or a combination thereof. The system **100** can include one or more database servers, in which each database server can include one, two, three, or another suitable number of databases configured as or comprising a suitable database type or combination thereof.

[0029] In some implementations, one or more databases, tables, other suitable information sources, or portions or combinations thereof may be stored, managed, or otherwise provided by one or more of the elements of the system **100** other than the database server **110**, for example, at least one of the clients **104A** through **104D** or the application server **108**.

[0030] The telephony server **112** enables network-based telephony and web communications from and to clients of a customer, such as the clients **104A** through **104B** for the customer **102A** or the clients **104C** through **104D** for the customer **102B**. Some or all of the clients **104A** through **104D** may be voice over internet protocol (VOIP)-enabled devices configured to send and receive calls over a network **114**. In particular, the telephony server **112** includes a session initiation protocol (SIP) zone and a web zone. The SIP zone enables a client of a customer, such as the customer **102A** or **102B**, to send and receive calls over the network **114** using SIP requests and responses. The web zone integrates telephony data with the application server **108** to enable telephony-based traffic access to software services run by the application server **108**. Given the combined functionality of the SIP zone and the web zone, the telephony server **112** may be or include a cloud-based private branch exchange (PBX) system.

[0031] The SIP zone receives telephony traffic from a client of a customer and directs same to a destination device. The SIP zone may include one or more call switches for routing the telephony traffic. For example, to route a VOIP call from a first VOIP-enabled client of a customer to a

second VOIP-enabled client of the same customer, the telephony server **112** may initiate a SIP transaction between a first client and the second client using a PBX for the customer. However, in another example, to route a VOIP call from a VOIP-enabled client of a customer to a client or non-client device (e.g., a desktop phone which is not configured for VOIP communication) which is not VOIP-enabled, the telephony server **112** may initiate a SIP transaction via a VOIP gateway that transmits the SIP signal to a public switched telephone network (PSTN) system for outbound communication to the non-VOIP-enabled client or non-client phone. Hence, the telephony server **112** may include a PSTN system and may in some cases access an external PSTN system.

[0032] The telephony server **112** includes one or more session border controllers (SBCs) for interfacing the SIP zone with one or more aspects external to the telephony server **112**. In particular, an SBC can act as an intermediary to transmit and receive SIP requests and responses between clients or non-client devices of a given customer with clients or non-client devices external to that customer. When incoming telephony traffic for delivery to a client of a customer, such as one of the clients **104A** through **104D**, originating from outside the telephony server **112** is received, a SBC receives the traffic and forwards it to a call switch for routing to the client.

[0033] In some implementations, the telephony server **112**, via the SIP zone, may enable one or more forms of peering to a carrier or customer premise. For example, Internet peering to a customer premise may be enabled to ease the migration of the customer from a legacy provider to a service provider operating the telephony server **112**. In another example, private peering to a customer premise may be enabled to leverage a private connection terminating at one end at the telephony server **112** and at the other end at a computing aspect of the customer environment. In yet another example, carrier peering may be enabled to leverage a connection of a peered carrier to the telephony server **112**.

[0034] In some such implementations, a SBC or telephony gateway within the customer environment may operate as an intermediary between the SBC of the telephony server **112** and a PSTN for a peered carrier. When an external SBC is first registered with the telephony server **112**, a call from a client can be routed through the SBC to a load balancer of the SIP zone, which directs the traffic to a call switch of the telephony server **112**. Thereafter, the SBC may be configured to communicate directly with the call switch.

[0035] The web zone receives telephony traffic from a client of a customer, via the SIP zone, and directs same to the application server **108** via one or more Domain Name System (DNS) resolutions. For example, a first DNS within the web zone may process a request received via the SIP zone and then deliver the processed request to a web service which connects to a second DNS at or otherwise associated with the application server **108**. Once the second DNS resolves the request, it is delivered to the destination service at the application server **108**. The web zone may also include a database for authenticating access to a software application for telephony traffic processed within the SIP zone, for example, a softphone.

[0036] The clients **104A** through **104D** communicate with the servers **108** through **112** of the datacenter **106** via the network **114**. The network **114** can be or include, for example, the Internet, a local area network (LAN), a wide area network (WAN), a virtual private network (VPN), or another public or private means of electronic computer communication capable of transferring data between a client and one or more servers. In some implementations, a client can connect to the network **114** via a communal connection point, link, or path, or using a distinct connection point, link, or path. For example, a connection point, link, or path can be wired, wireless, use other communications technologies, or a combination thereof.

[0037] The network **114**, the datacenter **106**, or another element, or combination of elements, of the system **100** can include network hardware such as routers, switches, other network devices, or combinations thereof. For example, the datacenter **106** can include a load balancer **116** for routing traffic from the network **114** to various servers associated with the datacenter **106**. The load balancer **116** can route, or direct, computing communications traffic, such as signals or messages,

to respective elements of the datacenter **106**.

[0038] For example, the load balancer **116** can operate as a proxy, or reverse proxy, for a service, such as a service provided to one or more remote clients, such as one or more of the clients **104A** through **104D**, by the application server **108**, the telephony server **112**, and/or another server. Routing functions of the load balancer **116** can be configured directly or via a DNS. The load balancer **116** can coordinate requests from remote clients and can simplify client access by masking the internal configuration of the datacenter **106** from the remote clients.

[0039] In some implementations, the load balancer **116** can operate as a firewall, allowing or preventing communications based on configuration settings. Although the load balancer **116** is depicted in FIG. **1** as being within the datacenter **106**, in some implementations, the load balancer **116** can instead be located outside of the datacenter **106**, for example, when providing global routing for multiple datacenters. In some implementations, load balancers can be included both within and outside of the datacenter **106**. In some implementations, the load balancer **116** can be omitted.

[0040] FIG. **2** is a block diagram of an example internal configuration of a computing device **200** of an electronic computing and communications system. In one configuration, the computing device **200** may implement one or more of one or more of the clients **104A** through **104D**, the application server **108**, the database server **110**, or the telephony server **112** of the system **100** shown in FIG. **1**.

[0041] The computing device **200** includes components or units, such as a processor **202**, a memory **204**, a bus **206**, a power source **208**, peripherals **210**, a user interface **212**, a network interface **214**, other suitable components, or a combination thereof. One or more of the memory **204**, the power source **208**, the peripherals **210**, the user interface **212**, or the network interface **214** can communicate with the processor **202** via the bus **206**.

[0042] The processor **202** is a central processing unit, such as a microprocessor, and can include single or multiple processors having single or multiple processing cores. Alternatively, the processor **202** can include another type of device, or multiple devices, configured for manipulating or processing information. For example, the processor **202** can include multiple processors interconnected in one or more manners, including hardwired or networked. The operations of the processor **202** can be distributed across multiple devices or units that can be coupled directly or across a local area or other suitable type of network. The processor **202** can include a cache, or cache memory, for local storage of operating data or instructions.

[0043] The memory **204** includes one or more memory components, which may each be volatile memory or non-volatile memory. For example, the volatile memory can be random access memory (RAM) (e.g., a DRAM module, such as DDR SDRAM). In another example, the non-volatile memory of the memory **204** can be a disk drive, a solid state drive, flash memory, or phase-change memory. In some implementations, the memory **204** can be distributed across multiple devices. For example, the memory **204** can include network-based memory or memory in multiple clients or servers performing the operations of those multiple devices.

[0044] The memory **204** can include data for immediate access by the processor **202**. For example, the memory **204** can include executable instructions **216**, application data **218**, and an operating system **220**. The executable instructions **216** can include one or more application programs, which can be loaded or copied, in whole or in part, from non-volatile memory to volatile memory to be executed by the processor **202**. For example, the executable instructions **216** can include instructions for performing some or all of the techniques of this disclosure. The application data **218** can include user data, database data (e.g., database catalogs or dictionaries), or the like. In some implementations, the application data **218** can include functional programs, such as a web browser, a web server, a database server, another program, or a combination thereof. The operating system **220** can be, for example, Microsoft Windows®, Mac OS X®, or Linux®; an operating system for a mobile device, such as a smartphone or tablet device; or an operating system for a

non-mobile device, such as a mainframe computer.

[0045] The power source **208** provides power to the computing device **200**. For example, the power source **208** can be an interface to an external power distribution system. In another example, the power source **208** can be a battery, such as where the computing device **200** is a mobile device or is otherwise configured to operate independently of an external power distribution system. In some implementations, the computing device **200** may include or otherwise use multiple power sources. In some such implementations, the power source **208** can be a backup battery.

[0046] The peripherals **210** includes one or more sensors, detectors, or other devices configured for monitoring the computing device **200** or the environment around the computing device **200**. For example, the peripherals **210** can include a geolocation component, such as a global positioning system location unit. In another example, the peripherals can include a temperature sensor for measuring temperatures of components of the computing device **200**, such as the processor **202**. In some implementations, the computing device **200** can omit the peripherals **210**.

[0047] The user interface **212** includes one or more input interfaces and/or output interfaces. An input interface may, for example, be a positional input device, such as a mouse, touchpad, touchscreen, or the like; a keyboard; or another suitable human or machine interface device. An output interface may, for example, be a display, such as a liquid crystal display, a cathode-ray tube, a light emitting diode display, or other suitable display.

[0048] The network interface **214** provides a connection or link to a network (e.g., the network **114** shown in FIG. **1**). The network interface **214** can be a wired network interface or a wireless network interface. The computing device **200** can communicate with other devices via the network interface **214** using one or more network protocols, such as using Ethernet, transmission control protocol (TCP), internet protocol (IP), power line communication, an IEEE 802.X protocol (e.g., Wi-Fi, Bluetooth, or ZigBee), infrared, visible light, general packet radio service (GPRS), global system for mobile communications (GSM), code-division multiple access (CDMA), Z-Wave, another protocol, or a combination thereof.

[0049] FIG. **3** is a block diagram of an example of a software platform **300** implemented by an electronic computing and communications system, for example, the system **100** shown in FIG. **1**. The software platform **300** is a UCaaS platform accessible by clients of a customer of a UCaaS platform provider, for example, the clients **104A** through **104B** of the customer **102A** or the clients **104C** through **104D** of the customer **102B** shown in FIG. **1**. The software platform **300** may be a multi-tenant platform instantiated using one or more servers at one or more datacenters including, for example, the application server **108**, the database server **110**, and the telephony server **112** of the datacenter **106** shown in FIG. **1**.

[0050] The software platform **300** includes software services accessible using one or more clients. For example, a customer **302** as shown includes four clients **304** through **310** (e.g., the clients **304**, **306**, **308**, **310**)—a desk phone, a computer, a mobile device, and a shared device. The desk phone is a desktop unit configured to at least send and receive calls and includes an input device for receiving a telephone number or extension to dial to and an output device for outputting audio and/or video for a call in progress. The computer is a desktop, laptop, or tablet computer including an input device for receiving some form of user input and an output device for outputting information in an audio and/or visual format. The mobile device is a smartphone, wearable device, or other mobile computing aspect including an input device for receiving some form of user input and an output device for outputting information in an audio and/or visual format. The desk phone, the computer, and the mobile device may generally be considered personal devices configured for use by a single user. The shared device is a desk phone, a computer, a mobile device, or a different device which may instead be configured for use by multiple specified or unspecified users.

[0051] Each of the clients **304** through **310** includes or runs on a computing device configured to access at least a portion of the software platform **300**. In some implementations, the customer **302** may include additional clients not shown. For example, the customer **302** may include multiple

clients of one or more client types (e.g., multiple desk phones or multiple computers) and/or one or more clients of a client type not shown in FIG. 3 (e.g., wearable devices or televisions other than as shared devices). For example, the customer **302** may have tens or hundreds of desk phones, computers, mobile devices, and/or shared devices.

[0052] The software services of the software platform **300** generally relate to communications tools, but are in no way limited in scope. As shown, the software services of the software platform **300** include telephony software **312**, conferencing software **314**, messaging software **316**, and other software **318**. Some or all of the software **312** through **318** uses customer configurations **320** specific to the customer **302**. The customer configurations **320** may, for example, be data stored within a database or other data store at a database server, such as the database server **110** shown in FIG. 1.

[0053] The telephony software **312** enables telephony traffic between ones of the clients **304** through **310** and other telephony-enabled devices, which may be other ones of the clients **304** through **310**, other VOIP-enabled clients of the customer **302**, non-VOIP-enabled devices of the customer **302**, VOIP-enabled clients of another customer, non-VOIP-enabled devices of another customer, or other VOIP-enabled clients or non-VOIP-enabled devices. Calls sent or received using the telephony software **312** may, for example, amongst the clients **304** through **310** be sent or received using the desk phone, a softphone running on the computer, a mobile application running on the mobile device, or using the shared device that includes telephony features.

[0054] The telephony software **312** further enables phones that do not include a client application to connect to other software services of the software platform **300**. For example, the telephony software **312** may receive and process calls from phones not associated with the customer **302** to route that telephony traffic to one or more of the conferencing software **314**, the messaging software **316**, or the other software **318**.

[0055] The conferencing software **314** enables audio, video, and/or other forms of conferences between multiple participants, such as to facilitate a conference between those participants. In some cases, the participants may all be physically present within a single location, for example, a conference room, in which the conferencing software **314** may facilitate a conference between only those participants and using one or more clients within the conference room. In some cases, one or more participants may be physically present within a single location and one or more other participants may be remote, in which the conferencing software **314** may facilitate a conference between all of those participants using one or more clients within the conference room and one or more remote clients. In some cases, the participants may all be remote, in which the conferencing software **314** may facilitate a conference between the participants using different clients for the participants. The conferencing software **314** can include functionality for hosting, presenting scheduling, joining, or otherwise participating in a conference. The conferencing software **314** may further include functionality for recording some or all of a conference and/or documenting a transcript for the conference.

[0056] The messaging software **316** enables instant messaging, unified messaging, and other types of messaging communications between multiple devices, such as to facilitate a chat or other virtual conversation between users of those devices. The unified messaging functionality of the messaging software **316** may, for example, refer to email messaging which includes a voicemail transcription service delivered in email format.

[0057] The other software **318** enables other functionality of the software platform **300**. Examples of the other software **318** include, but are not limited to, device management software, resource provisioning and deployment software, administrative software, third party integration software, and the like. In one particular example, the other software **318** can include software for training an intent matching engine of a contact center.

[0058] The software **312** through **318** may be implemented using one or more servers, for example, of a datacenter such as the datacenter **106** shown in FIG. 1. For example, one or more of the

software **312** through **318** may be implemented using an application server, a database server, and/or a telephony server, such as the servers **108** through **112** shown in FIG. **1**. In another example, one or more of the software **312** through **318** may be implemented using servers not shown in FIG. **1**, for example, a meeting server, a web server, or another server. In yet another example, one or more of the software **312** through **318** may be implemented using one or more of the servers **108** through **112** and one or more other servers. The software **312** through **318** may be implemented by different servers or by the same server.

[0059] Features of the software services of the software platform **300** may be integrated with one another to provide a unified experience for users. For example, the messaging software **316** may include a user interface element configured to initiate a call with another user of the customer **302**. In another example, the telephony software **312** may include functionality for elevating a telephone call to a conference. In yet another example, the conferencing software **314** may include functionality for sending and receiving instant messages between participants and/or other users of the customer **302**. In yet another example, the conferencing software **314** may include functionality for file sharing between participants and/or other users of the customer **302**. In some implementations, some or all of the software **312** through **318** may be combined into a single software application run on clients of the customer, such as one or more of the clients **304** through **310**.

[0060] FIG. **4** is a block diagram of an example of a contact center system. A contact center **400**, which in some cases may be implemented in connection with a software platform (e.g., the software platform **300** shown in FIG. **3**), is accessed by a user device **402** and used to establish a connection between the user device **402** and an agent device **404** over one of multiple modalities available for use with the contact center **400**, for example, telephony, video, text messaging, chat, and social media. The contact center **400** is implemented using one or more servers and software running thereon. For example, the contact center **400** may be implemented using one or more of the servers **108** through **112** shown in FIG. **1**, and may use communication software such as or similar to the software **312** through **318** shown in FIG. **3**. The contact center **400** includes software for facilitating contact center engagements requested by user devices such as the user device **402**. As shown, the software includes request processing software **406**, agent selection software **408**, and session handling software **410**.

[0061] The request processing software **406** processes a request for a contact center engagement initiated by the user device **402** to determine information associated with the request. The request may include a natural language query or a request entered in another manner (e.g., “press 1 to pay a bill, press 2 to request service”). The information associated with the request generally includes information identifying the purpose of the request and which is usable to direct the request traffic to a contact center agent capable of addressing the request. The information associated with the request may include information obtained from a user of the user device **402** after the request is initiated. For example, for the telephony modality, the request processing software **406** may use an interactive voice response (IVR) menu to prompt the user of the user device **402** to present information associated with the purpose of the request, such as by identifying a category or sub-category of support requested. In another example, for the video modality, the request processing software **406** may use a form or other interactive user interface to prompt a user of the user device **402** to select options which correspond to the purpose of the request. In yet another example, for the chat modality, the request processing software **406** may ask the user of the user device **402** to summarize the purpose of the request (e.g., the natural language query) via text and thereafter process the text entered by the user device **402** using natural language processing and/or other processing.

[0062] The session handling software **410** establishes a connection between the user device **402** and the agent device **404**, which is the device of the agent selected by the agent selection software **408**. The particular manner of the connection and the process for establishing same may be based

on the modality used for the contact center engagement requested by the user device **402**. The contact center engagement is then facilitated over the established connection. For example, facilitating the contact center engagement over the established connection can include enabling the user of the user device **402** and the selected agent associated with the agent device **404** to engage in a discussion over the subject modality to address the purpose of the request from the user device **402**. The facilitation of the contact center engagement over the established connection can use communication software implemented in connection with a software platform, for example, one of the software **312** through **318**, or like software.

[0063] The user device **402** is a device configured to initiate a request for a contact center engagement which may be obtained and processed using the request processing software **406**. In some cases, the user device **402** may be a client device, for example, one of the clients **304** through **310** shown in FIG. 3. For example, the user device **402** may use a client application running thereat to initiate the request for the contact center engagement. In another example, the connection between the user device **402** and the agent device **404** may be established using software available to a client application running at the user device **402**. Alternatively, in some cases, the user device **402** may be other than a client device.

[0064] The agent device **404** is a device configured for use by a contact center agent. Where the contact center agent is a human, the agent device **404** is a device having a user interface. In some such cases, the agent device **404** may be a client device, for example, one of the clients **304** through **310**, or a non-client device. In some such cases, the agent device **404** may be a server which implements software usable by one or more contact center agents to address contact center engagements requested by contact center users. Where the contact center agent is a non-human, the agent device **404** is a device that may or may not have a user interface. For example, in some such cases, the agent device **404** may be a server which implements software of or otherwise usable in connection with the contact center **400**.

[0065] Although the request processing software **406**, the agent selection software **408**, and the session handling software **410** are shown as separate software components, in some implementations, some or all of the request processing software **406**, the agent selection software **408**, and the session handling software **410** may be combined. For example, the contact center **400** may be or include a single software component which performs the functionality of all of the request processing software **406**, the agent selection software **408**, and the session handling software **410**. In some implementations, one or more of the request processing software **406**, the agent selection software **408**, or the session handling software **410** may be comprised of multiple software components. In some implementations, the contact center **400** may include software components other than the request processing software **406**, the agent selection software **408**, and the session handling software **410**, such as in addition to or in place of one or more of the request processing software **406**, the agent selection software **408**, and the session handling software **410**.

[0066] FIG. 5 is a block diagram of an example of a system **500** for training an intent matching engine. As shown, the system **500** includes user devices **502A-C**, a server **504**, and an administrator device **506**. Each of the user devices **502A-C** may correspond to the user device **402**. The server **504** may be associated with the contact center **400** and may perform at least one of the request processing software **406**, the agent selection software **408**, or the session handling software **410** of the contact center **400**. The administrator device **506** may correspond to the agent device **404** or may be distinct from the agent device **404**. Thus, the administrator device **506** may be operated by an administrator, who may or may not be an agent.

[0067] As shown, the server stores queries **508A-C** received from the user devices **502A-C**, with the query **508A** being received from the user device **502A**, the query **508B** being received from the user device **502B**, and the query **508C** being received from the user device **502C**. More details of examples of data stored in conjunction with each query **502A-C** are provided in conjunction with FIG. 6. The queries **508A-C** are provided to the intent matching engine **510**. The intent matching

engine **510** may be implemented using one or more artificial intelligence techniques. For example, the intent matching engine **510** may include an artificial neural network (e.g., a convolutional neural network and/or a deep neural network). The intent matching engine **510** computes, for each of the queries **508A-C**, a confidence score representing a probability that the query matches to an intent (or each of multiple intents from a set of intents). The confidence score may be computed using the artificial neural network based on a feature vector including text, audio, and/or video of the query.

[0068] As shown, based on the confidence score, the query **508A** is matched, by the intent matching engine **510**, to the intent **512**. For example, the query **508A** may be matched to the intent **512** due to the confidence score of the query **508A** being matched to the intent **512** exceeding a matching threshold (e.g., as described in conjunction with FIG. 7). In one example, the query **508A** may include the text “I would like to close my account,” and the intent **512** may be “close account.”

[0069] The query **508B** and the query **508C** are not matched to any intents, for example, due to their confidence scores not exceeding the matching threshold. As a result, the query **508B** and the query **508C** are provided, by the intent matching engine **510**, to a selection engine **514** for processing.

[0070] The selection engine **514** identifies queries to be manually matched to an intent **516** by a user of the administrator device **506** in order to generate training data **518** for online training of the intent matching engine **510**. In some cases, at most a predetermined number (e.g., three or five) of queries may be provided to the administrator device **506**, so as not to overwhelm the user of the administrator device **506**. In some cases, the queries provided to the administrator device **506** may have a timestamp at most a threshold time (e.g., one week) before a current time to ensure that the training data **518** includes recently generated queries, as queries from user devices **502A-C** may change over time based on changes in the circumstances or the needs of the users of the user devices **502A-C**. In some cases, the queries provided to the administrator device **506** may be queries having a confidence score within a range. The range may be between an uncertainty threshold and the matching threshold. In one example, confidence scores range between 0 and 1. The uncertainty threshold is 0.7 and the matching threshold is 0.8. As a result of these values, queries that are “almost” matched to the intent are used for training the intent matching engine **510** so as to increase the confidence scores of those queries if those queries are repeated or similar queries are presented.

[0071] As illustrated, the selection engine **514** selects the query **508B** (but not the query **508C**) for provision to the administrator device **506**. At the administrator device **506**, the user provides an input for matching the query **508B** to the intent **516**. The query **508B** and intent **516** pair is transmitted to the server **504**, where the pair is used as training data **518** for online learning-based training of the intent matching engine **510**. As illustrated, a single query-intent match is made by the administrator device **506**. However, in some implementations, multiple query-intent pairs may be provided, by the administrator device **506**, to the training data **518**. In some cases, multiple administrator devices (including the administrator device **506**) may be used. The multiple administrator devices **506** may receive mutually exclusive queries to match to intents.

Alternatively, a single query may be provided to multiple administrator devices for matching to an intent and may be stored in the training data **518** if at least a threshold number (e.g., at least two or at least 50% of the total number of administrator devices to which the query was provided) match the query to the same intent.

[0072] Online-learning based training of the intent matching engine **510** may include training the intent matching engine **510** after it has been used for inference in the real world. For example, the intent matching engine **510** may be used to match queries to intents and, after such use, be provided with additional training data to refine the query-intent matching. After the additional training data is used to train the intent matching engine **510**, the intent matching engine is once again used for

inference. This may be similar to how a human initially learns a skill (e.g., how to bake a cake) and then refines their skill based on feedback they receive in the real world. For example, a person might learn that their cake is not sufficiently sweet and may add more sugar in the next iteration.

[0073] FIG. 6 is a block diagram of an example of stored data associated with a query 600. The query 600 may correspond to one of the queries 508A-C. As shown, the query 600 stores a query audio 602. The query audio 602 may be an audio or video recording of speech by a user at the user device 502A-C in response to being asked to state a reason for accessing the contact center 400. For example, the query audio 602 may include an audio or video of a user saying, “I would like to return the product.” The query audio 602 may be converted to query text 604 using a speech-to-text engine (e.g., an artificial neural network trained to convert speech to text in a natural language) residing at the server 504 or another machine. The query text may store the text generated by the speech-to-text engine, for example, the text, “I would like to return the product.” In some cases, the user may type the query 600 (e.g., if the user is using or has access to a chat interface) instead of speaking the query to an audio or video recorder. In these cases, there would be no query audio 602 and the typed text would be stored as the query text 604. It should be noted that the user is notified that they are being recorded and persistent notifications (e.g., a visual icon) may be presented at the user device 502A-C to inform the user that the recording is ongoing. The user may opt out of being recorded or of having their recording stored after the contact center engagement is completed. In some cases, if the user is using video calling technology to access the contact center, the user may agree to have their audio recorded while disagreeing to have visual data from the user device 502A-C recorded, as the query text 604 may be generated without using visual data.

[0074] As shown, the query 600 has a timestamp 606. The timestamp 606 represents a time when the user spoke the query audio 602 or typed the query text 604. The query 600 has a confidence score 608. The confidence score 608 represents a likelihood that the query 600 matches to an intent, as calculated by the intent matching engine 510. In some cases, multiple confidence scores 608 may be calculated for matching the query 600 to multiple different intents. For example, the query “I would like a replacement for the new version of the product,” may have a confidence score of 0.75 for the intent “exchange product,” a confidence score of 0.63 the intent “return product,” and a confidence score of 0.09 for the intent “close account.” In these cases, the highest confidence score (e.g., the confidence score of 0.75 for the intent “replace product”) may be used as the confidence score 608, as the query is most likely to match to the intent associated with the highest confidence score.

[0075] As illustrated, a training value score 610 is computed based on the timestamp 606 and the confidence score 608. The training value score 610 may be used, by the selection engine 514, to determine whether to transmit the query 600 to the administrator device 506 for manual intent matching to generate the training data 518. The training value score 610 may be computed to favor more recent timestamps and confidence scores that represent “almost” being matched to an intent. In one example, the training value score 610 is a Boolean value that is set to true if the confidence score is greater than 80% of the matching threshold and less than the matching threshold and if the timestamp is within one month of the current time, and is set to false otherwise. The selection engine 514 may select, for provision to the administrator device 506, queries having a training value score of true.

[0076] FIG. 7 is a number line 700 of values of confidence scores (e.g., the confidence score 608) associated with a query matching to an intent. As shown, the confidence score values range from 0.0 to 1.0, and include an uncertainty threshold 702 and a matching threshold 704. Confidence scores below the uncertainty threshold 702 correspond to a not match range 706. The not match range 706 indicates that the query is very unlikely to match to the intent. Thus, adding queries associated with the not match range 706 to the training data 518 might not be very valuable.

[0077] Confidence scores between the uncertainty threshold 702 and the matching threshold 704 correspond to an uncertainty range 708. A query falling in the uncertainty range 708 is not

sufficiently likely to match to the intent for the intent matching engine **510** to assign the query to the intent but is “close enough” to the matching threshold **704** that receiving confirmation (e.g., via the administrator device **506**) of whether the query matches to the intent and training the intent matching engine **510** based on the confirmation may be useful.

[0078] Confidence scores exceeding the matching threshold **704** correspond to a match range **710**. A query falling in the match range **710** is matched to the associated intent by the intent matching engine **510**, and the associated user device is then processed according to a workflow for the intent and/or assigned to an agent who is capable of handling the intent. The workflow may include a series of manual or automated steps to perform to handle the intent. For example, for the intent “return product,” the workflow may include (1) identifying the product and the order, (2) verifying that the product is eligible for returning, and (3) generating a shipping label to return the product.

[0079] The uncertainty threshold **702** and/or the matching threshold **704** may correspond to different numeric values. In one example, the uncertainty threshold **702** is equal to 0.5, and the matching threshold **704** is equal to 0.8. The not match range **706** corresponds to 0.0 through 0.5. the uncertainty range **708** correspond to 0.5 through 0.8. The match range **710** corresponds to 0.8 through 1.0. In other examples, different values of the uncertainty threshold **702** and/or the matching threshold **704** may be used. Furthermore, the bounds of the number line **700** are shown as 0.0 and 1.0; however, in alternative implementations, confidence scores may be bounded by values different from 0.0 and 1.0, for example, confidence scores may be expressed in the interval of −100 through 100.

[0080] FIG. **8** is an example GUI **800** for matching intents to queries at a client device. As shown, the GUI **800** is presented at a mobile device **802** (e.g., a mobile phone or a tablet computer). However, the GUI **800** may be presented at another computing device, such as a laptop computer or a desktop computer. The GUI **800** may be presented at a client device capable of communicating (e.g., via a network) with the server **504**, such as the administrator device **506**.

[0081] As illustrated, the mobile device **802** includes a screen **804** that presents queries for review. The screen **804** includes query representations **806A-C**. As shown, the query representation **806A** includes a query text (e.g., the query text **604**)—“I don't like the product I got as a gift. Can I exchange it for money?”—and a menu **808A** for selecting an intent matching to the query. The menu **808A**, when selected (e.g., by touching the menu **808A** with a finger or a stylus), may include menu items for different intents to which the query may be matched (e.g., all available intents or intents for which the confidence score **608** exceeds the uncertainty threshold **702**). Using the menu **808A**, the user of the mobile device **802** may select a correct intent to which the query represented by the query representation **806A** is to be matched, generating a query-intent pair that is added to the training data **518** of the intent matching engine **510**. Similarly, the query representation **806B** includes a query text—“Can I get a replacement instead of my money back?”—and a menu **808B**. The query representation **806C** includes a query text—“Return a defective product using the warranty.”—and the menu **808C**.

[0082] To further describe some implementations in greater detail, reference is next made to examples of techniques which may be performed by a contact center server in online learning-based training an intent matching engine and inference using the intent matching engine. FIG. **9** is a flowchart of an example of a technique **900** for training an intent matching engine of a contact center. The technique **900** can be executed using computing devices, such as the systems, hardware, and software described with respect to FIGS. **1-8**. The technique **900** can be performed, for example, by executing a machine-readable program or other computer-executable instructions, such as routines, instructions, programs, or other code. The steps, or operations, of the technique **900** or another technique, method, process, or algorithm described in connection with the implementations disclosed herein can be implemented directly in hardware, firmware, software executed by hardware, circuitry, or a combination thereof.

[0083] For simplicity of explanation, the technique **900** is depicted and described herein as a series

of steps or operations. However, the steps or operations in accordance with this disclosure can occur in various orders and/or concurrently. Additionally, other steps or operations not presented and described herein may be used. Furthermore, not all illustrated steps or operations may be required to implement a technique in accordance with the disclosed subject matter.

[0084] The technique **900** may be implemented at a server of a contact center, for example, the server **504** or a server of the contact center **400**. The technique **900** may be implemented in parallel with at least one of the request processing software **406**, the agent selection software **408**, or the session handling software **410**.

[0085] At **902**, the server of the contact center accesses queries (e.g., the queries **508A-C**) provided by user devices (e.g., the user devices **502A-C**). The server may convert the received queries into a standardized format. For example, the queries may be converted to text and foreign language queries may be translated into a default natural language (e.g., English).

[0086] At **904**, the server determines a confidence scores (e.g., the confidence score **608**) for queries matching to intents. Each confidence score represents a likelihood that a query of the queries matches an intent. The confidence scores are determined using an intent matching engine (e.g., the intent matching engine **510**). The confidence score may be determined using statistical or machine learning techniques. The likelihood may be a probability or another representation of likelihood (e.g., an odds ratio).

[0087] At **906**, the server matches the queries to the intents based on the confidence scores exceeding a threshold (e.g., the matching threshold **704**) and using the intent matching engine. Using the confidence scores, some queries are successfully matched to an intent, while other queries are not matched to any intent (e.g., due to the confidence score being below the threshold). If the intent is matched to the query, the user device that submitted the query may be processed based on the intent. For example, the user device may be connected to an agent device with an agent that is capable of handling the intent. Alternatively, an automated workflow may be used to handle the intent. For example, if user wishes to purchase a product, the user may be provided with a web address where they can make the purchase or the user may be prompted, by an automated engine at the server, to provide their payment information and shipping information.

[0088] At **908**, the server provides, to a client device (e.g., the administrator device **506**), a subset of the queries that are not matched to any intent. The server identifies the subset based on timestamps or confidence scores. In some cases, the subset includes queries having timestamps during a threshold time period (e.g., within one week of a current time) and confidence scores within a range (e.g., between the uncertainty threshold **702** and the matching threshold **704**). In some cases, the subset is identified based on a training utility score (e.g., the training value score **610**) of each query. The training utility score may be computed based on a combination (e.g., a mathematical or logical combination) of the timestamp and the confidence score.

[0089] In some cases, all of the queries in the subset are provided to a single client device for the user of the single client device to match the queries to intents. Alternatively, if multiple client devices are available, the queries in the subset may be divided among the multiple client devices to reduce the workload provided to each client device user. For example, if there are ten client devices available and fifty queries in the subset, five queries may be provided to each client device to avoid overwhelming the users of the client devices.

[0090] At **910**, the server receives, from the client device (or from one or more of the multiple client devices), data representing intents for the subset of queries. In some cases, the user of the client device may fail to match at least one query to an intent, and the server may receive data representing an intent that matches at least one query in the subset. The server may cause the client device to display a GUI (e.g., as shown in FIG. **8**) indicating the subset of queries and a prompt, to the user of the client device, to match the queries to intents. When the user matches the queries to the intents, the server may receive, from the client device, a notification of the query-intent matches or a notification that a given query does not match to any intent.

[0091] At **912**, the server trains the intent matching engine based on the received data. A supervised learning technique may be used, as the received data includes queries labeled with intents. Alternatively, another machine learning technique, for example semi-supervised learning or reinforcement learning may be used. The intent matching engine may be trained using online learning, as the training may take place after the intent matching engine is already being used for inference in the real world.

[0092] The training of the intent matching engine may occur in real-time when the data is received. Alternatively, the training may be scheduled to occur during off peak hours when the contact center is predicted to be accessed by fewer than a predefined number of user devices or when the load on the contact center server is predicted to be (or is actually) below a threshold. For example, the training may be scheduled to occur between the hours 2:00 AM and 4:00 AM, when few users are expected to access the contact center. If there are unusual circumstances and multiple users actually access the contact center during those hours, the training may be rescheduled until the volume of users accessing the contact center decreases.

[0093] Some implementations are described below as numbered examples (Example 1, 2, 3, etc.). These examples are provided as examples only and do not limit the other implementations disclosed herein.

[0094] Example 1 is a method comprising: accessing queries provided by user devices to a contact center server; determining a confidence score representing a likelihood that a query of the queries matches an intent using an intent matching engine; matching, via the intent matching engine, the query to the intent when the confidence score exceeds a threshold; providing, to a client device, a subset of the queries that are not matched to any intent, the subset being identified based on confidence scores; receiving, from the client device, data representing an intent that matches at least one query in the subset; and training the intent matching engine based on the received data from the client device.

[0095] In Example 2, the subject matter of Example 1 includes, the subset comprising a predefined number of queries and the subset comprising queries having the confidence score within a range.

[0096] In Example 3, the subject matter of Examples 1-2 includes, the intent matching engine comprising a deep convolutional neural network trained via online learning.

[0097] In Example 4, the subject matter of Examples 1-3 includes, providing different subsets of queries to each of multiple different client devices, the client device being one of the multiple different client devices; receiving the data from the multiple client devices; and training the intent matching engine based on the received data from the multiple client devices.

[0098] In Example 5, the subject matter of Examples 1-4 includes, wherein the training of the intent matching engine occurs during off peak hours when the contact center server is accessed by fewer than a predefined number of user devices.

[0099] In Example 6, the subject matter of Examples 1-5 includes, wherein providing, to the client device, the subset of queries comprises: causing display, at the client device, of a graphical user interface indicating at least one of the subset of queries, a collection of intents, or a prompt to match the queries to the intents.

[0100] In Example 7, the subject matter of Examples 1-6 includes, wherein accessing the queries comprises: accessing at least one of an audio query or a video query; and converting the at least one of the audio query or the video query to a text query using a speech-to-text engine.

[0101] In Example 8, the subject matter of Examples 1-7 includes, identifying, using a selection engine, the subset of queries for provision to the client device, the selection engine calculating a training value score for a query from the subset of queries based on the confidence score and determining, based on the training value score, whether to place the query into the subset of queries.

[0102] Example 9 is a non-transitory computer readable medium storing instructions operable to cause one or more processors to perform operations comprising: accessing queries provided by user

devices to a contact center server; determining a confidence score representing a likelihood that a query of the queries matches an intent using an intent matching engine; matching, via the intent matching engine, the query to the intent when the confidence score exceeds a threshold; providing, to a client device, a subset of the queries that are not matched to any intent, the subset being identified based on confidence scores; receiving, from the client device, data representing an intent that matches at least one query in the subset; and training the intent matching engine based on the received data from the client device.

[0103] In Example 10, the subject matter of Example 9 includes, the subset comprising queries having the confidence score within a range.

[0104] In Example 11, the subject matter of Examples 9-10 includes, the intent matching engine comprising a deep neural network trained via online supervised learning.

[0105] In Example 12, the subject matter of Examples 9-11 includes, the operations comprising: providing subsets of queries to each of multiple client devices, the client device being one of the multiple client devices; and receiving the data from the multiple client devices; and training the intent matching engine based on the received data from the multiple client devices.

[0106] In Example 13, the subject matter of Examples 9-12 includes, wherein the training of the intent matching engine occurs during a time when the contact center server is accessed by fewer than a predefined number of user devices.

[0107] In Example 14, the subject matter of Examples 9-13 includes, wherein providing, to the client device, the subset of queries comprises: causing presentation, at the client device, of a user interface indicating the subset of queries or a collection of intents.

[0108] In Example 15, the subject matter of Examples 9-14 includes, wherein accessing the queries comprises: accessing at least one query comprising audio; and converting the at least one query comprising audio to a text query using a speech-to-text engine.

[0109] In Example 16, the subject matter of Examples 9-15 includes, the operations comprising: identifying the subset of queries for provision to the client device by calculating a training value score for a query from the subset of queries based on the confidence score and determining, based on the training value score, whether to place the query into the subset of queries.

[0110] Example 17 is an apparatus comprising: a memory; and a processor configured to execute instructions stored in the memory to: access queries provided by user devices to a contact center server; determine a confidence score representing a likelihood that a query of the queries matches an intent using an intent matching engine; match, via the intent matching engine, the query to the intent when the confidence score exceeds a threshold; provide, to a client device, a subset of the queries that are not matched to any intent, the subset being identified based on confidence scores; receive, from the client device, data representing an intent that matches at least one query in the subset; and train the intent matching engine based on the received data from the client device.

[0111] In Example 18, the subject matter of Example 17 includes, the subset comprising queries having the confidence score within a range.

[0112] In Example 19, the subject matter of Examples 17-18 includes, the intent matching engine comprising a convolutional neural network trained via online learning.

[0113] In Example 20, the subject matter of Examples 17-19 includes, wherein the training of the intent matching engine is scheduled to occur when the contact center server is accessed by fewer than a predefined number of user devices.

[0114] Example 21 is at least one machine-readable medium including instructions that, when executed by processing circuitry, cause the processing circuitry to perform operations to implement of any of Examples 1-20.

[0115] Example 22 is an apparatus comprising means to implement of any of Examples 1-20.

[0116] Example 23 is a system to implement of any of Examples 1-20.

[0117] Example 24 is a method to implement of any of Examples 1-20.

[0118] The implementations of this disclosure can be described in terms of functional block

components and various processing operations. Such functional block components can be realized by a number of hardware or software components that perform the specified functions. For example, the disclosed implementations can employ various integrated circuit components (e.g., memory elements, processing elements, logic elements, look-up tables, and the like), which can carry out a variety of functions under the control of one or more microprocessors or other control devices. Similarly, where the elements of the disclosed implementations are implemented using software programming or software elements, the systems and techniques can be implemented with a programming or scripting language, such as C, C++, Java, JavaScript, assembler, or the like, with the various algorithms being implemented with a combination of data structures, objects, processes, routines, or other programming elements.

[0119] Functional aspects can be implemented in algorithms that execute on one or more processors. Furthermore, the implementations of the systems and techniques disclosed herein could employ a number of conventional techniques for electronics configuration, signal processing or control, data processing, and the like. The words “mechanism” and “component” are used broadly and are not limited to mechanical or physical implementations, but can include software routines in conjunction with processors, etc. Likewise, the terms “system” or “tool” as used herein and in the figures, but in any event based on their context, may be understood as corresponding to a functional unit implemented using software, hardware (e.g., an integrated circuit, such as an ASIC), or a combination of software and hardware. In certain contexts, such systems or mechanisms may be understood to be a processor-implemented software system or processor-implemented software mechanism that is part of or callable by an executable program, which may itself be wholly or partly composed of such linked systems or mechanisms.

[0120] Implementations or portions of implementations of the above disclosure can take the form of a computer program product accessible from, for example, a computer-usable or computer-readable medium. A computer-usable or computer-readable medium can be a device that can, for example, tangibly contain, store, communicate, or transport a program or data structure for use by or in connection with a processor. The medium can be, for example, an electronic, magnetic, optical, electromagnetic, or semiconductor device.

[0121] Other suitable mediums are also available. Such computer-usable or computer-readable media can be referred to as non-transitory memory or media, and can include volatile memory or non-volatile memory that can change over time. The quality of memory or media being non-transitory refers to such memory or media storing data for some period of time or otherwise based on device power or a device power cycle. A memory of an apparatus described herein, unless otherwise specified, does not have to be physically contained by the apparatus, but is one that can be accessed remotely by the apparatus, and does not have to be contiguous with other memory that might be physically contained by the apparatus.

[0122] While the disclosure has been described in connection with certain implementations, it is to be understood that the disclosure is not to be limited to the disclosed implementations but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as is permitted under the law.

Claims

1. A method comprising: determining respective confidence scores for a query with respect to respective intents; providing, to a client device for review, an indication associated with the query based on a determination that the query does not match any of the respective intents based on the respective confidence scores; receiving, from the client device, data representing an intent of the respective intents that matches the query; and training an intent matching engine to associate the query with the intent based on the received data.

2. The method of claim 1, further comprising: converting an audio query to text using a speech-to-text engine; and determining the respective confidence scores based on the converted text.
3. The method of claim 1, wherein providing the indication associated with the query comprises: identifying that the query is within a confidence score range below a matching threshold; and causing the client device to display the indication.
4. The method of claim 1, further comprising: selecting a subset of queries not associated with an intent; and transmitting the subset to multiple client devices for parallel review.
5. The method of claim 1, wherein training the intent matching engine to associate the query with the intent based on the received data comprises: adding the query and the intent from the received data to training data; and executing a training operation to refine matching performance.
6. The method of claim 1, further comprising: calculating a training value score for the query based on the respective confidence scores and a timestamp of the query; and selecting the query for provision to the client device based on the training value score.
7. The method of claim 1, wherein determining respective confidence scores comprises: processing the query using an artificial neural network to generate a feature vector; and computing the respective confidence scores based on the feature vector.
8. A system comprising: one or more memories; and one or more processors configured to execute instructions stored in the one or more memories to: determine respective confidence scores for a query with respect to respective intents; provide, to a client device for review, an indication associated with the query based on a determination that the query does not match any of the respective intents based on the respective confidence scores; receive, from the client device, data representing an intent of the respective intents that matches the query; and train an intent matching engine to associate the query with the intent based on the received data.
9. The system of claim 8, wherein the received data includes a selection of the intent from a user interface displaying a collection of intents.
10. The system of claim 8, wherein, to determine the respective confidence scores, the one or more processors configured to execute instructions stored in the one or more memories to: compute multiple confidence scores for the query corresponding to different intents and selecting a highest confidence score.
11. The system of claim 8, wherein the one or more processors further configured to execute instructions stored in the one or more memories to: access the query as an audio query from a user device; and convert the audio query to a text query using a speech-to-text engine before determining the respective confidence scores.
12. The system of claim 8, wherein the one or more processors further configured to execute instructions stored in the one or more memories to: provide different subsets of not matched queries to multiple client devices, including the client device; and receive data representing intents for the subsets from the multiple client devices for training the intent matching engine.
13. The system of claim 8, wherein the one or more processors further configured to execute instructions stored in the one or more memories to: translate the query from a foreign language to a default natural language; and process the translated query to determine the respective confidence scores.
14. One or more non-transitory computer readable media storing instructions operable to cause one or more processors to perform operations comprising: determining respective confidence scores for a query with respect to respective intents; providing, to a client device for review, an indication associated with the query based on a determination that the query does not match any of the respective intents based on the respective confidence scores; receiving, from the client device, data representing an intent of the respective intents that matches the query; and training an intent matching engine to associate the query with the intent based on the received data.
15. The one or more non-transitory computer readable media of claim 14, wherein providing, to the client device for review, the indication associated with the query comprises: causing display, at the

client device, of a graphical user interface indicating the query and a collection of intents; and prompting a user of the client device to select an intent from the collection of intents for the query.

16. The one or more non-transitory computer readable media of claim 14, the operations further comprising: selecting a subset of queries, including the query, that are not matched to any intent based on the respective confidence scores being within a predefined range; and providing indications associated with the subset of queries to the client device for review.

17. The one or more non-transitory computer readable media of claim 14, wherein the query is selected for provision to the client device based on a timestamp of the query being within a threshold time period from a current time.

18. The one or more non-transitory computer readable media of claim 14, the operations further comprising: determining that the query does not match any of the respective intents when no respective confidence score exceeds a matching threshold; and assigning the query to a subset of queries for manual review based on at least one confidence score being within a range below the matching threshold.

19. The one or more non-transitory computer readable media of claim 14, the operations further comprising: dividing not matched queries among a plurality of client devices; and training the intent matching engine based on matches received from each device.

20. The one or more non-transitory computer readable media of claim 14, wherein determining respective confidence scores comprises: generating a feature vector based on text of the query; and computing a likelihood for each intent using the feature vector.
